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Abstract

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Chapter 1

Mathematical and Cryptographic Background

This chapter introduces the notation and mathematical foundations used throughout the thesis.

1.1 Finite Fields

Definition 1.1 (Prime Field). Let p be a prime number. The *prime field* $\mathbb{F}_p$ is the set $\{0, 1, \dots, p-1\}$ equipped with addition and multiplication modulo p .

The multiplicative group $\mathbb{F}_p^* = \mathbb{F}_p \setminus \{0\}$ is cyclic of order $p-1$. An element $g \in \mathbb{F}_p^*$ is called a *generator* if $\{g^0, g^1, \dots, g^{p-2}\} = \mathbb{F}_p^*$. For any $a \in \mathbb{F}_p^*$, the multiplicative inverse can be computed via Fermat's little theorem:

$$a^{-1} \equiv a^{p-2} \pmod{p}. \quad (1.1)$$

1.1.1 The BabyBear Field

The implementation described in this thesis operates over the BabyBear prime field, defined by

$$p = 2^{31} - 2^{27} + 1 = 2013\,265\,921. \quad (1.2)$$

Since $p < 2^{31}$, every field element fits in a 32-bit unsigned integer, and the product of two elements fits in a 64-bit integer.

The multiplicative group $\mathbb{F}_p^*$ has order $p-1 = 2^{27} \cdot 15$. The factor 2^{27} , the *two-adicity* of the field, means that $\mathbb{F}_p^*$ contains a cyclic subgroup of order 2^{27} , which is essential for the Number Theoretic Transform.

1.1.2 Roots of Unity

Definition 1.2 (Primitive Root of Unity). An element $\omega \in \mathbb{F}_p$ is a *primitive N -th root of unity* if $\omega^N = 1$ and $\omega^k \neq 1$ for all $0 < k < N$.

Given a primitive N -th root of unity ω , the set $H = \{1, \omega, \omega^2, \dots, \omega^{N-1}\}$ is a multiplicative subgroup of $\mathbb{F}_p^*$ of order N . Because BabyBear has two-adicity 27, subgroups

of order 2^k for any $k \leq 27$ exist and are generated from a fixed multiplicative generator g by taking $g^{(p-1)/2^k}$.

These subgroups serve as evaluation domains for the polynomials in the STARK protocol: the execution trace is evaluated on a subgroup H of order 2^h , and FRI operates on a larger domain.

1.1.3 Field Extensions

Definition 1.3 (Field Extension). A *field extension* $\mathbb{F}_{p^k}$ of degree k over $\mathbb{F}_p$ is the quotient ring $\mathbb{F}_p[X]/(f(X))$, where $f(X) \in \mathbb{F}_p[X]$ is an irreducible polynomial of degree k . Elements are polynomials of degree less than k with coefficients in $\mathbb{F}_p$.

In this work we use the *quartic extension* $\mathbb{F}_{p^4}$, constructed via the irreducible polynomial $X^4 - 11$. The constant 11 is chosen because it is not a fourth power in $\mathbb{F}_p$, which guarantees that $X^4 - 11$ is irreducible. An element of $\mathbb{F}_{p^4}$ is a tuple $(c_0, c_1, c_2, c_3) \in \mathbb{F}_p^4$ representing $c_0 + c_1X + c_2X^2 + c_3X^3$, with the reduction rule $X^4 = 11$.

1.2 Polynomials

A univariate polynomial over $\mathbb{F}_p$ of degree at most d is $f(X) = \sum_{i=0}^d c_i X^i$ with $c_i \in \mathbb{F}_p$. It can be represented by its *coefficient vector* $(c_0, \dots, c_d)$ or by its *evaluation vector* on a domain D . Converting between the two representations, and evaluating a polynomial at a point, are the fundamental operations of the prover.

1.2.1 The Number Theoretic Transform

The conversion between coefficient and evaluation representations is performed using the Number Theoretic Transform (NTT), the finite-field analogue of the Fast Fourier Transform. When the evaluation domain is a multiplicative subgroup $D = \{1, \omega, \dots, \omega^{N-1}\}$ with N a power of 2, the NTT computes all evaluations $f(\omega^i)$ in $O(N \log N)$ operations. The algorithm exploits the decomposition

$$f(X) = f_{\text{even}}(X^2) + X \cdot f_{\text{odd}}(X^2),$$

recursing on the even- and odd-indexed coefficients over the squared subgroup, which has half the size. The inverse transform runs the same routine with ω^{-1} and divides by N .

1.2.2 Low-Degree Extension

The *low-degree extension* (LDE) of a polynomial is its evaluation on a domain larger than its coefficient count. Given the n coefficients of f , the LDE with blowup factor b evaluates f on a coset of a subgroup of size $b \cdot n$. The resulting evaluation vector is the Reed-Solomon codeword that the prover commits to. In our implementation the low-degree extension is computed by padding the coefficients with zeros, applying a coset shift, and running the NTT on the enlarged domain [Sta21, Section 3.4].

1.2.3 Vanishing Polynomials

Definition 1.4 (Vanishing Polynomial [Sta21, Section 5.1]). Given a set $S \subseteq \mathbb{F}$, the *vanishing polynomial* of S is $Z_S(X) := \prod_{s \in S} (X - s)$. When S is a multiplicative subgroup $H = \{1, \omega, \dots, \omega^{N-1}\}$, this simplifies to $Z_H(X) = X^N - 1$.

Z_H vanishes on every element of H and nowhere else. If a constraint polynomial $c(X)$ must be zero on H , then $Z_H(X)$ divides $c(X)$, and the quotient $q(X) = c(X)/Z_H(X)$ is a polynomial. The verifier checks that q has low degree, which certifies that c vanishes on H [Sta21]. This mechanism is the basis of the STARK construction presented in Chapter ??.

1.3 Coding Theory

Reed-Solomon codes, and distance from them, are the objects the FRI protocol operates on. We recall the general definitions that the soundness analysis of the FRI protocol relies on.

Definition 1.5 (Error-Correcting Code [Arn+24, Definition 3.1]). An *error-correcting code* of length n over an alphabet Σ is a subset $C \subseteq \Sigma^n$. The code is a *linear code* if $\Sigma = \mathbb{F}$ is a field and C is a linear subspace of $\mathbb{F}^n$. The *rate* of a linear code is $\rho := \dim(C)/n$.

The elements of C are its *codewords*, and the *minimum distance* of a code is the smallest relative Hamming distance (definition 1.8) between two distinct codewords. When a word lies far from every codeword, several codewords may still lie within a given distance of it; the following notion quantifies this.

Definition 1.6 (List Decoding [Arn+24, Definition 3.2]). For a code $C \subseteq \Sigma^n$, a word $f \in \Sigma^n$, and a proximity parameter $\delta \in [0, 1]$, the *list* of C at f within radius δ is

$$\Lambda(C, f, \delta) := \{u \in C : \Delta(f, u) \leq \delta\}.$$

The code is (δ, ℓ) -*list decodable* if $|\Lambda(C, f, \delta)| \leq \ell$ for every $f \in \Sigma^n$. A radius δ is *within unique decoding* if the code is $(\delta, 1)$ -list decodable.

For a code of minimum distance δ_C , any radius $\delta \leq \delta_C/2$ is within unique decoding. Beyond it, the list size for Reed-Solomon codes is controlled by the Johnson bound.

Theorem 1.1 (Johnson Bound [Arn+24, Theorem 4.3]). *The Reed-Solomon code of rate ρ is $(1 - \sqrt{\rho} - \eta, \frac{1}{2\eta\sqrt{\rho}})$ -list decodable for every $\eta \in (0, 1 - \sqrt{\rho})$.*

The two thresholds — the *unique decoding* radius $\frac{1-\rho}{2}$ and the *Johnson* radius $1 - \sqrt{\rho}$ — are exactly where the soundness guarantees of FRI change form (chapter 2).

1.4 Reed-Solomon Codes

Definition 1.7 (Reed-Solomon Code [BS+18, Section 2.1]). For a finite field $\mathbb{F}$, a subset $S \subseteq \mathbb{F}$, and a rate parameter $\rho \in (0, 1)$, the *Reed-Solomon code* $\text{RS}[\mathbb{F}, S, \rho]$ is the set of functions $f : S \rightarrow \mathbb{F}$ that are evaluations of polynomials of degree less than $\rho|S|$:

$$\text{RS}[\mathbb{F}, S, \rho] = \{(f(x))_{x \in S} : f \in \mathbb{F}[X], \deg(f) < \rho|S|\}.$$

Each codeword is the evaluation of a polynomial of degree less than $\rho|S|$ on all points of S . The code has minimum distance $|S| - \rho|S| + 1$, so two distinct codewords differ on a large fraction of positions.

The parameter ρ is the *rate*, and $1/\rho$ is the *blowup factor*. In our implementation the blowup factor is 2 ($\rho = 1/2$): the codeword produced by the low-degree extension is twice the length of the coefficient vector.

1.4.1 Relative Hamming Distance

Definition 1.8 (Relative Hamming Distance). The *relative Hamming distance* between two vectors $u, v \in \mathbb{F}^n$ is

$$\Delta(u, v) = \frac{|\{i : u_i \neq v_i\}|}{n}.$$

The *distance* of a vector u from a code C is $\Delta(u, C) = \min_{v \in C} \Delta(u, v)$. A vector u is δ -close to C if $\Delta(u, C) \leq \delta$, and δ -far if $\Delta(u, C) > \delta$.

The FRI protocol tests proximity to a Reed-Solomon codeword. As stated in [BS+18, Section 2.1], the verifier's task is to distinguish between the case that $f \in \text{RS}[\mathbb{F}, S, \rho]$ and the case that f is far from $\text{RS}[\mathbb{F}, S, \rho]$ in relative Hamming distance.

1.5 Cryptographic Hash Functions

Definition 1.9 (Collision-Resistant Hash Function). A *collision-resistant hash function* is a function $H : \{0, 1\}^* \rightarrow \{0, 1\}^\lambda$ such that no efficient algorithm can find distinct inputs $x \neq x'$ with $H(x) = H(x')$, except with negligible probability.

Hash functions serve two roles in STARK systems: constructing Merkle trees for committing to polynomial evaluations, and deriving the verifier's challenges via the Fiat-Shamir heuristic. Our implementation uses two different hash functions for these two roles: SHA-256 for the Merkle commitments and Keccak- $f[1600]$ for the challenger.

The distinction matters for the Fiat-Shamir transformation. Chiesa and Orrù [CO25] prove that a *duplex sponge* construction (such as Keccak/SHA-3) is suitable for Fiat-Shamir, while Merkle-Damgård constructions (such as SHA-256) lack a comparable security proof in this setting. For this reason we use a Keccak-based sponge for the challenger, while retaining SHA-256 for the Merkle trees, where only collision resistance is required.

1.6 Merkle Trees

Definition 1.10 (Merkle Tree). A *Merkle tree* over a sequence $(v_0, v_1, \dots, v_{N-1})$ with $N = 2^k$ is a binary tree where each leaf stores the hash of one element, $\ell_i = H(v_i)$; each internal node stores the hash of the concatenation of its two children, $H(n_{\text{left}} \| n_{\text{right}})$; and the root r is a single digest that binds the entire sequence.

To prove that v_i is the value at position i , the prover provides an *authentication path*: the $\log_2(N)$ sibling hashes along the path from the leaf to the root. The verifier recomputes the root and accepts if it matches r . Verification requires $O(\log N)$ hash evaluations.

In the BCS transformation [BSCS16], Merkle trees commit to the oracle messages of the IOP: the prover evaluates a polynomial on the FRI domain, builds a Merkle tree over the evaluations, and sends the root to the verifier.

1.7 Proof Systems

We now establish the formal framework of proof systems, following Thaler [Tha22].

1.7.1 Interactive Proofs

Definition 1.11 (Interactive Proof [Tha22, Definition 3.1]). An *interactive proof system* for a function f is a pair $(\mathcal{P}, \mathcal{V})$ where $\mathcal{P}$ is a computationally unbounded prover and $\mathcal{V}$ is a probabilistic polynomial-time verifier. The protocol proceeds in rounds, and must satisfy:

- **Completeness.** For every input x , $\Pr[\mathcal{V} \text{ accepts}] \geq 1 - \delta_c$.
- **Soundness.** For every input x and every cheating prover $\mathcal{P}^*$ that sends a value $y \neq f(x)$, $\Pr[\mathcal{V} \text{ accepts}] \leq \delta_s$.

1.7.2 Interactive Arguments

Relaxing the soundness of an interactive proof (definition 1.11) to hold only against efficient provers yields a weaker but more flexible notion.

Definition 1.12 (Interactive Argument [Tha22, Section 3.2]). An *interactive argument* for a language $\mathcal{L}$ is an interactive proof in which the soundness condition is only required to hold against prover strategies that run in probabilistic polynomial time.

Both models share the same structure. In a *public-coin* protocol, every verifier message is a uniformly random challenge, independent of the prover's messages, and the verifier's decision is a deterministic function of the *transcript* — the ordered sequence of messages and challenges. These are exactly the protocols to which the Fiat-Shamir transformation (section 1.7.3) applies. An *argument of knowledge* further requires an efficient extractor $\mathcal{E}$ that recovers a valid witness from any convincing prover.

STARKs are *arguments*, not *proofs* in the formal sense: their soundness relies on the collision resistance of a hash function, which a computationally unbounded prover could

break. The interactive oracle proofs of section 1.8 specialise this model by providing the prover’s messages as oracles.

1.7.3 The Fiat-Shamir Transformation

A *non-interactive* proof consists of a single message from the prover to the verifier. The Fiat-Shamir transformation removes interaction from a public-coin protocol by replacing the verifier’s random challenges with the output of a hash function applied to the transcript so far. Under the random oracle model, this preserves soundness. Chiesa and Orrù [CO25] give a formal treatment using a duplex sponge, which resolves the ambiguity between writing to and reading from the transcript by using the sponge’s native *absorb* and *squeeze* operations. Our challenger implements this construction over Keccak- $f[1600]$.

1.8 Interactive Oracle Proofs

Definition 1.13 (Interactive Oracle Proof [BSCS16, Definition 2.2]). An *Interactive Oracle Proof* (IOP) for a language $\mathcal{L}$ is an interactive protocol between a prover $\mathcal{P}$ and a verifier $\mathcal{V}$ that proceeds in rounds. In each round:

1. the prover sends an *oracle message* π_i (a function that the verifier may query at chosen positions);
2. the verifier sends a random challenge r_i .

After k rounds, the verifier makes a bounded number of queries to $\pi_1, \dots, \pi_k$ and outputs *accept* or *reject*.

The key difference from a standard interactive proof is that the verifier has *oracle access* to the prover’s messages: it reads only a small number of positions rather than the full message. This is what enables succinct verification.

An IOP of *proximity* (IOPP) is an IOP in which the verifier tests whether the prover’s initial oracle is close to a codeword of a specified code, rather than testing membership in a language [BS+18].

1.9 The BCS Transformation

The BCS transformation [BSCS16] compiles a public-coin IOP into a non-interactive argument in the random oracle model:

1. **Commitment.** Each oracle message is committed using a Merkle tree; the prover sends the root.
2. **Challenge derivation.** Each challenge is derived from the transcript via Fiat-Shamir.
3. **Query answering.** The prover opens the requested positions together with their Merkle authentication paths.

4. **Verification.** The verifier checks the paths, recomputes the challenges, and runs the IOP decision procedure.

This is precisely the structure of our FRI commit phase: each folding layer is committed with a Merkle tree, its root is absorbed into the challenger, and the next folding challenge is squeezed from it. The query and verification steps are performed when the verifier opens the Merkle paths and checks folding consistency.

1.10 Polynomial Commitment Schemes

Definition 1.14 (Polynomial Commitment Scheme). A *Polynomial Commitment Scheme* (PCS) consists of four algorithms:

- $\text{Setup}(\lambda, d) \rightarrow \text{pp}$: generates public parameters for polynomials of degree at most d .
- $\text{Commit}(\text{pp}, f) \rightarrow C$: produces a commitment C to the polynomial f .
- $\text{Open}(\text{pp}, f, z) \rightarrow (v, \pi)$: produces a proof π that $f(z) = v$.
- $\text{Verify}(\text{pp}, C, z, v, \pi) \rightarrow \{0, 1\}$: checks the evaluation proof.

The scheme must satisfy correctness and binding.

The STARK system implemented in this thesis uses a *FRI-based PCS* [BS+18]: a polynomial is committed via a Merkle tree over its low-degree extension, and evaluation proofs are constructed through the FRI low-degree testing protocol. This approach requires no trusted setup (transparent) and relies only on hash functions (post-quantum secure). The FRI protocol is described in detail in Chapter 2.

Chapter 2

The FRI Protocol

This chapter presents the Fast Reed-Solomon Interactive Oracle Proof of Proximity, the protocol at the core of the system implemented in this thesis. We begin by stating the problem FRI solves, then describe the algebraic structure of the evaluation domain, and finally the folding operation that drives the protocol.

2.1 Overview and Problem Statement

Recall from definition 1.7 that, for a subset $S \subseteq \mathbb{F}$ and a rate parameter $\rho \in (0, 1)$, the Reed-Solomon code $\text{RS}[\mathbb{F}, S, \rho]$ consists of the evaluations on S of polynomials of degree less than $\rho|S|$. The problem FRI addresses is the following.

Definition 2.1 (Rate Proximity Testing (RPT) [BS+18, Section 2.1]). A verifier is given oracle access to a function $f : S \rightarrow \mathbb{F}$, together with auxiliary information supplied by a prover. The verifier must distinguish, with high probability and using only a small number of queries, between the case that $f \in \text{RS}[\mathbb{F}, S, \rho]$ and the case that f is far from every member of $\text{RS}[\mathbb{F}, S, \rho]$ in relative Hamming distance (definition 1.8).

RPT is a special case of the low-degree testing problem, and Ben-Sasson et al. identify it as a major bottleneck for transparent proof systems [BS+18, Section 2.1].

A naive solution is instructive. The verifier could ask the prover to open f at $\rho|S|$ positions, interpolate the unique polynomial of degree less than $\rho|S|$ through them, and check that it agrees with f elsewhere. This fails on two counts: the number of openings is proportional to the degree bound, which may be in the millions, and every opening carries a Merkle authentication path of $\log |S|$ hash values (section 1.6), so the proof size becomes prohibitive.

FRI takes a different route. Instead of testing the degree directly, it *reduces* it: the prover repeatedly halves the degree of the committed polynomial using randomness supplied by the verifier, until a constant remains. Each reduction step is committed, and the verifier spot-checks the consistency of the steps at a small number of random positions. If the original function is far from the code, then with high probability at least one reduction is inconsistent, and the verifier detects it.

This yields a protocol with prover arithmetic complexity strictly linear in $|S|$ and verifier arithmetic complexity strictly logarithmic [BS+18, Section 2.1]. In the terminology of section 1.8, FRI is an interactive oracle proof of proximity (IOPP) for

Reed-Solomon codes: the prover's messages are committed codewords which the verifier queries at a bounded number of positions, rather than reading in full.

Remark 2.1. FRI establishes *proximity*, not membership. A successful verification does not certify that the committed vector is exactly a Reed-Solomon codeword, only that it is close to one in relative Hamming distance. This distinction is central to the soundness analysis of the FRI protocol.

2.1.1 Commit Phase

2.1.2 Query Phase

2.1.3 Verification and the Colinearity Check

2.2 Fiat-Shamir and the Challenger

2.2.1 The Transcript and its Ordering

2.2.2 The Duplex Sponge over Keccak

2.3 Soundness

2.3.1 Proximity, not Membership

2.3.2 Distance Preservation under Folding

2.3.3 Correlated Agreement and Mutual Correlated Agreement

2.3.4 Round-by-round Soundness of FRI

2.3.5 Provable versus Conjectured Regimes

2.4 Parameter Selection

Chapter 3

FRI as a Polynomial Commitment Scheme

3.1 From Low-Degree Testing to Evaluation Proofs

3.2 The Quotient Technique

3.3 Commit, Open, and Verify

3.4 Implementation Notes

Chapter 4

Conclusions

4.1 Summary

4.2 Limitations

4.3 Future Work

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